

Question 2: Client Churn Risk & Revenue Forecasting

Data Analytics Internship Assignment

1. Problem Statement

The company wants to identify clients who may churn in the next 3 months, understand the main churn factors, forecast revenue for the next 12 months and suggest ways to reduce churn by 5%.

2. Objectives

- Identify clients with high churn risk.
- Find the main churn factors.
- Forecast revenue for the next 12 months.
- Recommend actions to reduce churn.

3. Data Used and Cleaning

A sample of 200 enterprise clients was created because the original client dataset was not provided. It contains Monthly Revenue, Support Tickets, Satisfaction Score, Contract Months and Churn. Twelve percent of Monthly Revenue values (24 records) were made missing and filled using the median.

4. Churn Rate

Status	Clients	Percentage
No Churn	164	82%
Churn	36	18%

The sample churn rate is 18%, with 36 out of 200 clients classified as churned.

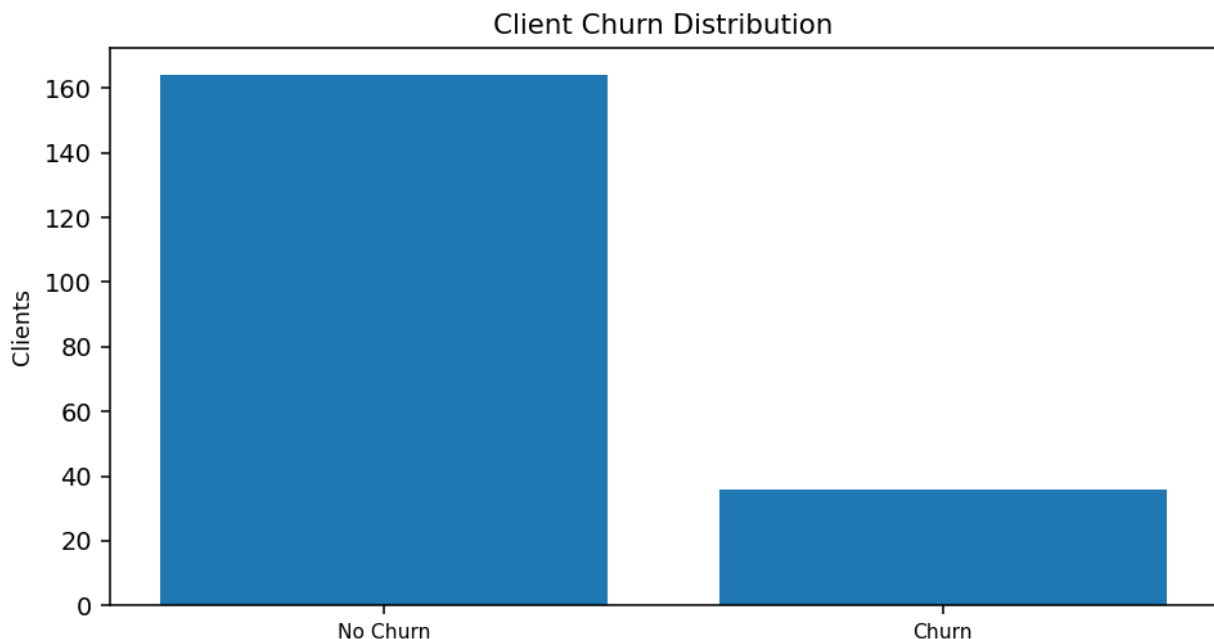


Figure 1: Client churn distribution.

5. Churn Risk Analysis

A simple churn-risk score was created using satisfaction, support-ticket volume and contract duration. Lower satisfaction, higher support-ticket volume and shorter contracts increase the risk score.

6. Main Churn Factors

Factor	Correlation with Churn
Satisfaction Score	-0.53
Support Tickets	+0.35
Monthly Revenue	+0.15
Contract Months	-0.10

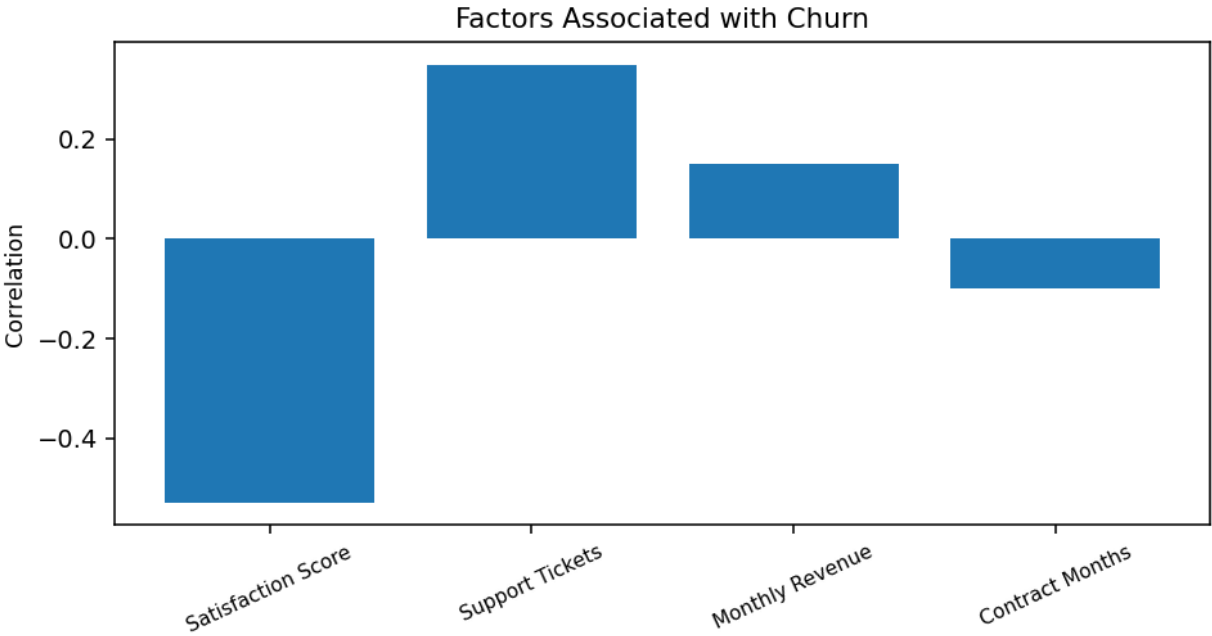


Figure 2: Correlation of selected factors with churn.

Satisfaction has the strongest relationship with churn in this sample. Support-ticket volume is the second important factor.

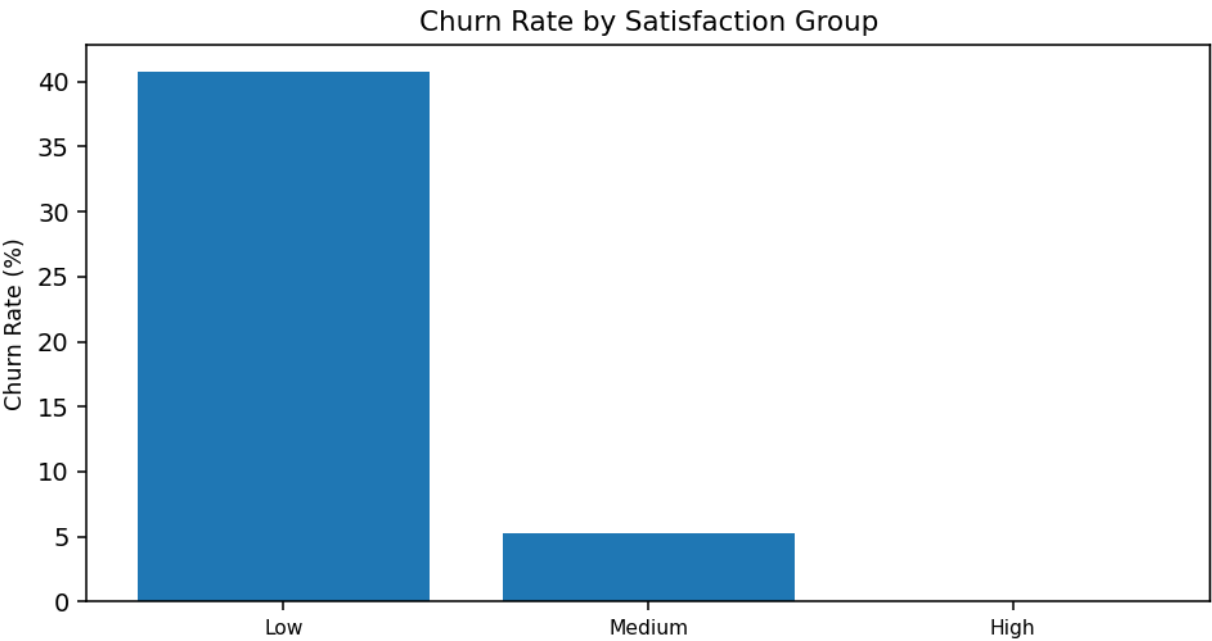


Figure 3: Churn rate by satisfaction group.

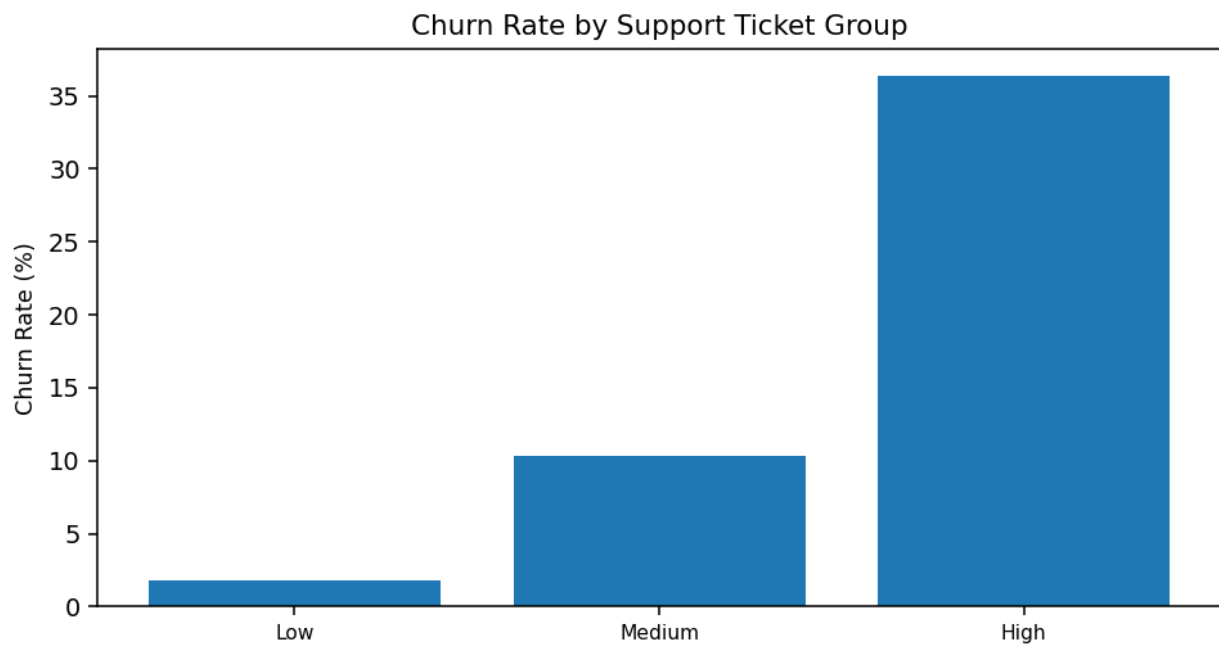


Figure 4: Churn rate by support-ticket group.

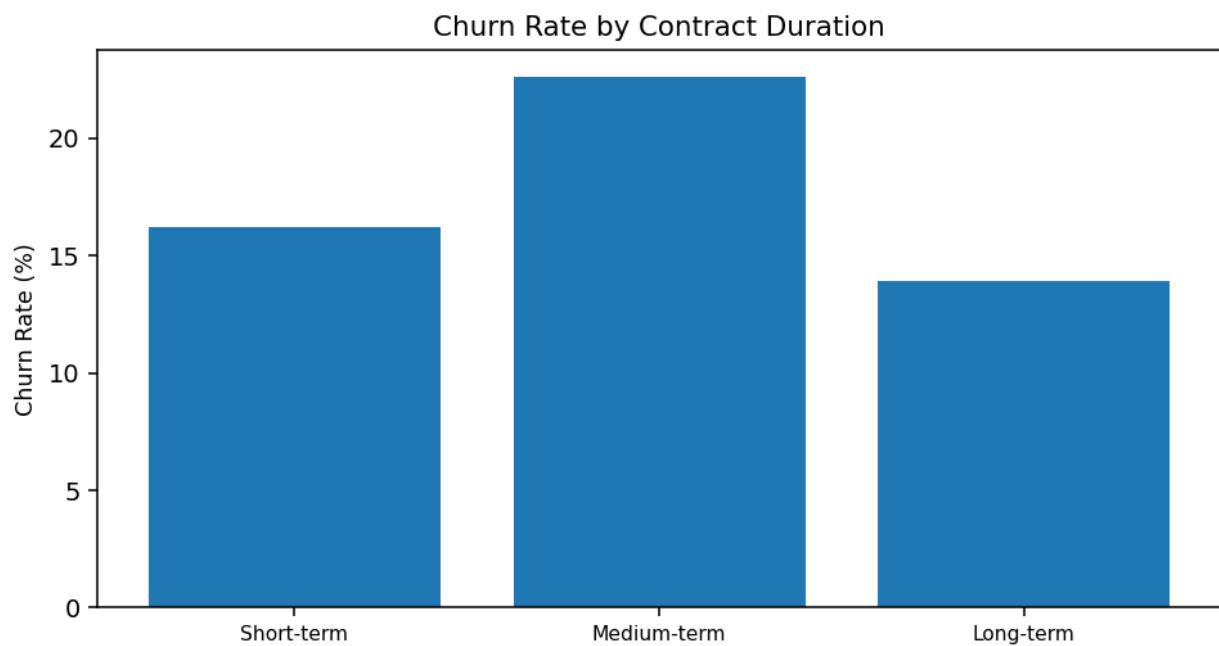


Figure 5: Churn rate by contract duration.

7. Revenue Forecasting

A 24-month sample revenue history was created for the 200 clients. Monthly revenue was aggregated and a simple linear trend was used to forecast the next 12 months.

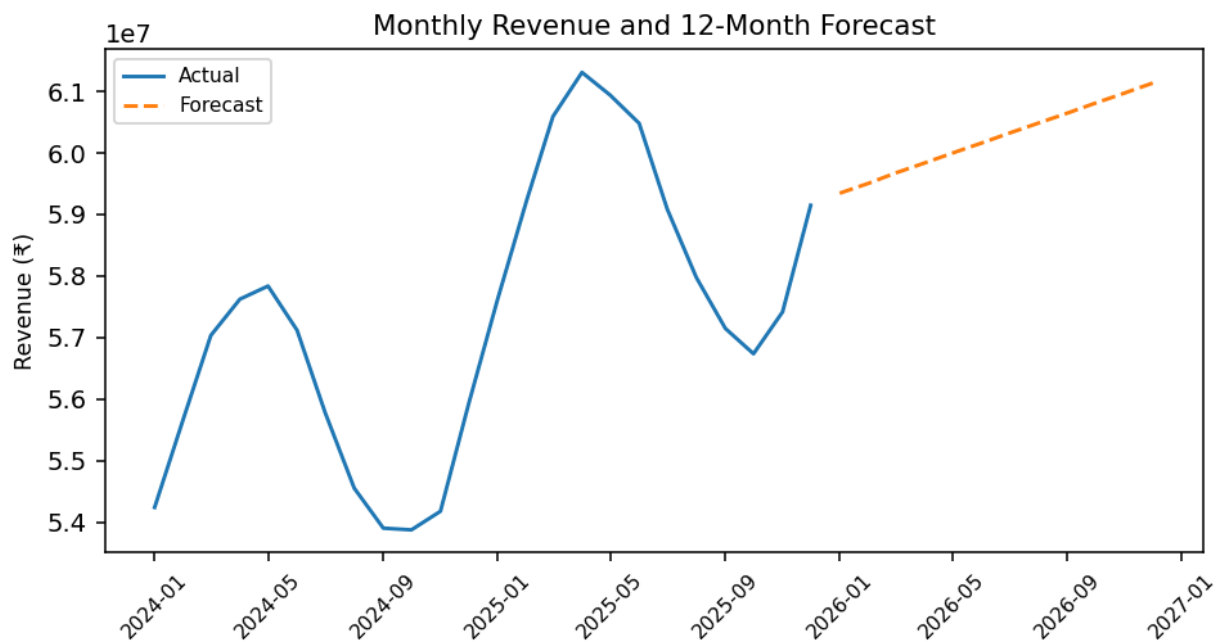


Figure 6: Monthly revenue trend and 12-month forecast.

Estimated 12-month forecast revenue: ■722,884,118.40 (approximately ■72.29 crore).

8. Recommendations

- Give proactive support to clients with low satisfaction.
- Identify and fix repeated support problems for clients with many tickets.
- Offer renewal benefits to short-term clients.
- Contact high-risk clients before contract renewal.
- Review customer health regularly using satisfaction, tickets, contract duration and revenue.

9. Key Findings

- Overall churn is 18%.
- Satisfaction is the strongest churn-related factor.
- Higher support-ticket volume is associated with higher churn risk.
- Shorter contracts generally have higher churn risk.
- A risk score can help prioritize retention efforts.

10. Conclusion

Improving customer satisfaction and support experience should be the main focus for reducing churn. High-risk clients can be targeted with proactive retention actions, while better renewals and service improvements can help protect recurring revenue.

Note: Sample/synthetic client and revenue data was used for this assignment.